TEAM AURA

CareFlow OPD

Outpatient Appointment Booking System

Project overview • MVP requirements • AI use • Next step

Software Engineering Project

1. Problem & Users

Who experiences the problem, and what do they need?

PATIENTS

Patients need a convenient way to:

- Find doctors and check availability
- Book, cancel or reschedule appointments
- Receive a token and check their own appointment status
- Manage their profile and notifications

HOSPITAL ADMIN STAFF

Admins need a clear way to:

- Maintain doctor information and schedules
- Monitor appointments, tokens and dashboard summaries
- Manage patient account access
- Review records without manually controlling the queue

Core problem: Appointment information and access need to be easier to manage for both patients and hospital staff.

2. Must-have Functional Requirements (MVP)

The nine functional requirements defined in the SRS

FR-1	Patients view doctor information and availability.
FR-2	Patients book appointments, provide optional visit reasons, and receive tokens.
FR-3	Patients view their own token numbers and appointment statuses.
FR-4	Patients cancel, reschedule, or arrange follow-up appointments.
FR-5	Admins manage doctor information and schedules.
FR-6	Admins monitor dashboards, appointments, and token records.
FR-7	Admins manage Patient Account Access.
FR-8	Patients register, sign in and manage profiles.
FR-9	Patients and Admins receive and manage notifications.

MVP focus: appointment booking and token information, supported by admin and account-management functions.

Testing Snapshot

Summary based on the team's observed acceptance-criteria checks

PASS

Doctor search, booking selection, token display, cancellation, rescheduling, admin management and account controls were observed.

PARTIAL PASS

Token uniqueness / duplicate prevention and the full range of notification behavior still need verification.

NOT TESTED

Follow-up booking was not tested because no follow-up appointment was needed during the test.

Examples of observed results

- Booked time slots could not be selected again.
- Rescheduling to an available slot generated a new token.
- Banned accounts showed an “account access is restricted” message.
- Patients could not view other patients’ personal tokens.

These are prototype test observations, not evidence of a production-ready hospital backend.

3. AI Use: Support, Review & Changes

ChatGPT was used as an assistant; the team remained responsible for decisions.

WHERE AI HELPED

- Brainstorming and clarifying the project idea
- Organizing SRS requirements and acceptance criteria
- Explaining UML and system flows
- Supporting HTML, CSS and JavaScript troubleshooting
- Improving documentation and presentation wording

WHAT THE TEAM CHECKED

- Requirement-to-screen consistency
- UML actors, boundaries and relationships
- Prototype pages, navigation and visible interactions
- File names, folder structure and repository contents
- Presentation claims against the actual prototype

WHAT THE TEAM CHANGED

- Kept only suggestions that matched the agreed scope
- Revised wording and mappings after team review
- Corrected confusing diagram elements
- Avoided claiming backend or production features that were not implemented

AI output can be inaccurate. The team checked, edited and took responsibility for all submitted work.

4. Next Step: Improve Booking Reliability

One improvement with a clear reason

PROPOSED IMPROVEMENT

Strengthen duplicate-booking prevention with a reliable booking process that checks slot availability when a booking is confirmed and assigns each successful booking a unique token.

WHY THIS MATTERS

It reduces the risk of two patients receiving the same appointment slot.

WHAT TO DO NEXT

Implement and verify the booking logic beyond the current front-end demonstration.

HOW TO VERIFY

Test two booking attempts for the same slot and confirm only one succeeds.

Thank you